Classification Assignment

Logistic Regression:

```
Fitting 5 folds for each of 4 candidates, totalling 20 fits
The f1_macro value for best parameter{'penalty': 'l2', 'solver': 'lbfgs'}: 0.9916844900066377
The confusion matrix:
 [[45  0]
 [ 1 74]]
              precision    recall  f1-score   support

           0       0.98      1.00      0.99        45
           1       1.00      0.99      0.99        75

    accuracy                           0.99       120
   macro avg       0.99      0.99      0.99       120
weighted avg       0.99      0.99      0.99       120
```

Support Vector Machine:

```
Fitting 5 folds for each of 24 candidates, totalling 120 fits
The f1_macro value for best parameter{'C': 10, 'gamma': 'auto', 'kernel': 'sigmoid'}: 0.9834018801410106
The confusion matrix:
 [[45  0]
 [ 2 73]]
              precision    recall  f1-score   support

           0       0.96      1.00      0.98        45
           1       1.00      0.97      0.99        75

    accuracy                           0.98       120
   macro avg       0.98      0.99      0.98       120
weighted avg       0.98      0.98      0.98       120
```

RandomForestClassifier:

```
The f1_macro value for best parameter{'criterion': 'entropy', 'max_features': 'log2', 'n_estimators': 100}: 0.9916844900066377
The confusion matrix:
 [[45  0]
 [ 1 74]]
              precision    recall  f1-score   support

           0       0.98      1.00      0.99        45
           1       1.00      0.99      0.99        75

    accuracy                           0.99       120
   macro avg       0.99      0.99      0.99       120
weighted avg       0.99      0.99      0.99       120
```

Classification Assignment

DecisionTreeClassifier:

```
Fitting 5 folds for each of 12 candidates, totalling 60 fits
The f1_macro value for best parameter{'criterion': 'gini', 'max_features': 'sqrt', 'splitter': 'random'}: 0.9668037602820213
The confusion matrix:
 [[44  1]
 [ 3 72]]
              precision    recall  f1-score   support

           0       0.94      0.98      0.96        45
           1       0.99      0.96      0.97        75

    accuracy                           0.97       120
   macro avg       0.96      0.97      0.96       120
weighted avg       0.97      0.97      0.97       120
```

Algorithm results

Logistic Regression: 0.99

Random Forest Classifier: 0.99

Decision Tree Classifier: 0.97

Support Vector Machine: 0.98

Best Algorithm's:

Logistic Regression

RandomForestClassifier